Rui Li

PhD Student · Aalto University

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Summary

- Ph.D. student at Aalto University, supervised by Prof. Arno Solin. Published on ICLR, ICML, and WACV.
- Research focuses on uncertainty quantification, probabilistic ML, and few-shot learning, with experience in LLMs and VLMs.
- Interned at Amazon Web Services (KV-cache compression) and Nokia Bell Labs (uncertainty for robotics foundation models).
- Proficient in Python and Pytorch. Developed SUQ library. Strong mathematical skills in linear algebra, calculus, and probability.

Experience

Aalto University, Ph.D. *2022 - Present*

- Advised by Prof. Arno Solin and Prof. Martin Trapp
- Research interests: uncertainty quantification, probabilistic machine learning, few-shot learning
- Final year: published on ICML, ICLR, WACV and four NeurIPS workshops

Amazon Web Services, Research Internship *Oct 2025 - Jan 2026*

- KV cache compression for long-context LLM with Dr. Matthias Seeger

Nokia Bell Labs, Research Internship *March 2025 - May 2025*

- Uncertainty quantification for robotics foundation models with Dr. Mikko Honkala

University College London, M.Sc. in Machine Learning, Distinction (84%), Dean's List *2020 - 2021*

- Thesis: Learning Input-conditional Invariances via the Marginal Likelihood
  Supervised by Prof. Marc Deisenroth, Dr. So Takao, and Prof. Mark van der Wilk

Tokyo University, Research Student *2019 - 2020*

- Researched aligning contextual word embedding in a probabilistic manner with variational autoencoder.
- Supervised by Prof. Yoshimasa Tsuruoka

Sun Yat-sen University, B.Sc. in Physics, GPA: 3.8 / 4.0 *2015 - 2019*

- Thesis: Deep Learning for Vertex Reconstruction in JUNO Experiment, supervised by Prof. Zhengyun You

Selected Publications

Streamlining Prediction in Bayesian Deep Learning. ***ICLR 2025***

Rui Li, Marcus Klasson, Arno Solin, Martin Trapp.

In BDL, how to make predictions efficiently is overlooked. We examine streamlining prediction in BDL through a single forward pass without sampling. We showcase our approach for both MLP and transformers, such as ViT and GPT-2.

Post-hoc Probabilistic Vision-Language Models. ***ICLR 2026***

Anton Baumann, Rui Li, Marcus Klasson, Santeri Mentu, Shyamgopal Karthik, Zeynep Akata, Arno Solin, Martin Trapp.

To make VLMs more reliable, we present the first direct Bayesian formulation. We propose a post-hoc uncertainty quantification method based on Laplace approximation, which provides useful predictive uncertainties and better calibration.

Flatness Improves Backbone Generalisation in Few-shot Classification. ***WACV (Oral) 2025***

Rui Li, Martin Trapp, Marcus Klasson, Arno Solin.

In few-shot classification most efforts focus on adapting the backbone to the target domain without considering the importance of backbone training. We show flatness-aware backbone training can lead to better generalization through theoretical and empirical results.

Improving Hyperparameter Learning under Approximate Inference in Gaussian Process Models. ***ICML 2023***

Rui Li, ST John, Arno Solin.

Variational inference and expectation propagation are two commonly used approximate inferences in Gaussian process models with complementary advantages. We developed a hybrid training procedure to bring the best of both worlds.

Programming Languages and Tech Skills

- Python, PyTorch, NumPy, TensorFlow, GPFlow, GitHub
- Developed SUQ, a lightweight PyTorch library for uncertainty quantification

I use PyTorch and GitHub on a daily basis for my research. I run experiments on GPU clusters such as LUMI, the fifth fastest supercomputer in the world.

Academic Service

- Organizer of workshop on Uncertainty Quantification for Computer Vision

  Together with colleagues, I co-organized the second and third workshops on Uncertainty Quantification for Computer Vision at ICCV 2023 and ECCV 2024, top venues for computer vision.

- Reviewed for NeurIPS, ICML, ICLR and ECCV workshop.

Awards

- Dean's list at University College London, 2021

  The Dean's List is awarded to the top 5% of graduating students.

- First, third, and second price scholarship at Sun Yat-sen University, 2019, 2018, 2017

  Scholarship is awarded every year based on grades and publications.

- Meritorious Winner of Interdisciplinary Contest in Modeling, 2017

  Organizer: COMAP.

Referees

- Prof. Arno Solin (arno.solin@aalto.fi), Arno is my Ph.D. supervisor.
- Prof. Martin Trapp (mtrapp@kth.se), I work with Martin during my Ph.D. on few-shot learning and Bayesian deep learning.
- Dr. ST John (ti.john@aalto.fi), I work with Ti during my Ph.D. on approximate inference in Gaussian processes.

Full List Publications

- Sophie Steger, Rui Li, Sofiane Ennadir, Anya Sims, Arno Solin, Martin Trapp, Franz Pernkopf. Stochasticity in Tokenization Improves Robustness. *Under Review*
- Anton Baumann, Rui Li, Marcus Klasson, Santeri Mentu, Shyamgopal Karthik, Zeynep Akata, Arno Solin, Martin Trapp. Post-hoc Probabilistic Vision-Language Models.. *International Conference on Learning Representations (ICLR)*, 2026.
- Rui Li, Marcus Klasson, Arno Solin, Martin Trapp. Streamlining Prediction in Bayesian Deep Learning. *International Conference on Learning Representations (ICLR)*, 2025.
- Rui Li, Martin Trapp, Marcus Klasson, Arno Solin. Flatness Improves Backbone Generalisation in Few-shot Classification. *Winter Conference on Applications of Computer Vision (WACV)*, Oral Presentation, 2025.
- Rui Li, Marcus Klasson, Arno Solin, Martin Trapp. Posterior Inferred, Now What? Streamlining Prediction in Bayesian Deep Learning. *NeurIPS Workshop on Bayesian Decision-making and Uncertainty* 2024.
- Anton Baumann, Marcus Klasson, Rui Li, Arno Solin, Martin Trapp. Probabilistic Active Few-Shot Learning in Vision-Language Models. *NeurIPS Workshop on Bayesian Decision-making and Uncertainty* 2024.
- Marlon Tobaben, Marcus Klasson, Rui Li, Arno Solin, Antti Honkela. Differentially Private Continual Learning using Pre-Trained Models. *NeurIPS workshop on Scalable Continual Learning for Lifelong Foundation Models* 2024.
- Rui Li, ST John, Arno Solin. Improving Hyperparameter Learning under Approximate Inference in Gaussian Process Models. *International Conference on Machine Learning (ICML)*, 2023.
- Rui Li, ST John, Arno Solin. Towards Improved Learning in Gaussian Processes: The Best of Two Worlds. *NeurIPS Workshop on Gaussian Processes, Spatiotemporal Modeling, and Decision-making Systems,* 2022.
- Arno Solin, Rui Li, Andrea Pilzer. A Look at Improving Robustness in Visual-inertial SLAM by Moment Matching. *International Conference on Information Fusion (FUSION)*, 2022.
- Chuan Chen, Rui Li, Lin Shu, Zhiyu He, Jining Wang, Chengming Zhang, Huanfei Ma, Kazuyuki Aihara, Luonan Chen. Predicting Future Dynamics From Short-term Time Series Using an Anticipated Learning Machine. *National Science Review,* 2020.
- Rui Li, Fanghua Ye, Shaoan Xie, Chuan Chen and Zibin Zheng. Digging into It: Community Detection via Hidden Attributes Analysis. *Neurocomputing,* 2019.

- Rui Li, Zhengyun You and Yumei Zhang. Deep Learning for Signal and Background Discrimination in Liquid based Neutrino Experiment. *International Workshop on Advanced Computing and Analysis Techniques in Physics Research (ACAT)*, 2018.